FFGFT (T0) in the IPI Field

A map: substrate and primitive axes

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Abstract

This document places FFGFT (T0, time–mass duality) within the field of frameworks discussed at the *Information Physics Institute* (IPI). Rather than another pairwise comparison (those exist in Doc. 244, 245/269, 246, 251, 260, 271, 272), it proposes a *map* along two orthogonal organizing axes: (1) **substrate geometry** – what the discrete underlay is made of – and (2) the **primitive** – what counts as fundamental. On axis 1 FFGFT occupies the *periodic* pole (a compact T^4 crystal), HLV the *aperiodic* one (a quasicrystal); the *random* pole remains a control model with no committed owner. This triad is exactly the one the spectral discriminator testbed (Doc. 271, accompanying script) renders measurable. On axis 2 FFGFT is geometry-primitive, alongside information-primitive (Vopson; downstream Leavitt), algebraic/Hilbert-space (UMF) and selection/constraint frameworks (RA/PMT, RSG, UC, UIFT). It is stated explicitly that this two-axis map is the *author's organizing overlay* – in Dot-Theory terms a *secondary rendering (O2)*, a declared reading – not the native vocabulary of the respective frameworks; the primary rendering (O0) remains with their originators.

.1 Purpose and limits of this map

Over recent months FFGFT has been set against a series of frameworks in the IPI exchange. Each comparison ended strikingly often with the same finding: *a category difference, not a contradiction*. This document asks why, and proposes a map that explains it.

Three caveats first. *One*: the labels below ("periodic crystal lattice", "information-primitive", etc.) are *organizing terms of this work*. Some originators would classify their framework differently; the map is a reading, not a verdict. *Two* – and this is the more important point: the *choice of the two axes* (substrate geometry, primitive) is itself a *declared projection*, not a universal dimension of the field. Another researcher could just as admissibly choose different axes – such as Accessibility/Constraint or Observer-Dependence/Recoverability – and arrive at a different but equally legitimate classification of the same field. The axes used here are therefore carried as a *separately identified projection*, not as implied universals; they are privileged only relative to the stated purpose of this map. *Three*: even within this projection the axes do not sort the field exhaustively – Axis 1 (substrate) cleanly separates only the geometric substrate programmes; algebraic (UMF) and continuum-topological (αLGQV) programmes sit beside the triad. That is not a defect of the map but its honest resolution.

Use of O0/O2 (declared): O0/O2 are used here in the *provenance-preserving* sense – markers of where interpretive authority resides and who retains the right of correction – not as a full import of Guevara's rendering hierarchy (O0/O1/O2 as representational levels). Whether the two ultimately coincide is left open; this map uses only the provenance-preserving reading (cf. Vossen's FAH concern). Rendering level and provenance coincide in this document only because a secondary rendering of *another's* framework is necessarily authored by someone other than its originator – outside cross-framework comparison they decouple.

.2 Axis 1 – substrate geometry (the crystal axis)

FFGFT is at heart a *periodic crystal lattice*: the compact 4-torus T^4 is periodic and ordered, with fractal correction $D_f = 3 - \xi$. HLV, by contrast, is an *aperiodic quasicrystal* (cut-and-project $\mathbb{Z}^5 \to 3$D, golden ratio / Fibonacci). A third, *random* substrate serves as a null model. These three types are not merely conceptually distinct – they have *different spectral fingerprints*, and it is precisely this separation that the graph-Laplacian testbed of Doc. 271 makes scale-free measurable.

Substrate type	Representative	Spectral fingerprint (scale-free)
periodic crystal lattice	**FFGFT** (T^4)	strongly degenerate; $\langle r\rangle \approx 0$; integer $\|n\|^2$ ladder (sums of squares); in $D{=}4$ *every* integer is hit (Lagrange); no Cantor gaps
aperiodic quasicrystal	**HLV** (Krüger)	low degeneracy; *intermediate* ("critical") $\langle r\rangle$; high gap richness (Cantor hierarchy) – expected, distinguishable from both controls
random substrate	control model	$\langle r\rangle \approx 0.50$ (GOE 0.531); no degeneracy, no gap hierarchy

The discriminators ($\langle r\rangle$ = consecutive level-spacing ratio, degeneracy fraction, gap richness) are dimensionless and need no unfolding; the absolute scale plays no role. Axis 1 is therefore not merely a picture but an *experiment*: the test of whether a given spectrum is periodic, aperiodic, or random. FFGFT contributes the periodic-arm prediction (the rational $|n|^2$ ladder), HLV the aperiodic substrate; the guardrail "$X = \varphi^N$ or $X = \xi^N$ is not a prediction" (Doc. 190 R35) holds for both.

.3 Axis 2 – the primitive (what everything follows from)

A large part of the IPI field has *no* geometric substrate; there the crystal axis does not apply, and a second organizing term is needed: what counts as fundamental?

- **Geometry-primitive: FFGFT** – everything follows from T^4 geometry and the single dimensionless constant $\xi \approx 1/7500$. Static, observer-free, no selector.
- **Information-primitive: Vopson** – mass-energy-information equivalence (MEI), second law of infodynamics, simulation hypothesis. Information is physical and fundamental. Downstream: **Leavitt**, whose bit-budget construction sits directly on Vopson's MEI.
- **Algebraic / Hilbert-space: UMF** (Gericke) – a prime-indexed Hilbert space $\ell^2(P_N)$ with coupling bivector B_{OI}.
- **Selection / constraint** (what *selects or constrains* the configuration): **RA/PMT** (Guevara), **RSG** (Austin), **UC** (Haskins), **UIFT** (Teker).
- **Continuum-topological:** αLGQV (Kriger) – S^3 topology, membrane elasticity as metric.
- **Meta / governance** (orders *other* theories, one level up): **Dot Theory** (Vossen).

On this axis the FFGFT contrast with Vopson is not periodic-vs-aperiodic but the *geometric versus the informational side of the same MEI relation*: FFGFT approaches from geometry, Vopson from entropy (Doc. 251). A visible structural difference follows – FFGFT structurally excludes an "outside" or a simulation (there is no external standpoint to a single static whole), whereas Vopson's framework leaves it open.

.4 The overall map

Framework	Owner	Category (axis)	Relation to FFGFT	Doc.
FFGFT / T0	Pascher	substrate: periodic; primitive: geometry	— (reference point: static, observer-free, no selector)	—
HLV	Krüger	substrate: aperiodic	shared spectral/recovery axis; otherwise a category mismatch	271, 272
Vopson (MEI)	Vopson	primitive: information	geometric vs. informational side of the same MEI relation; outside excluded vs. open	251
(bit budget)	Leavitt	primitive: information (downstream Vopson)	shares Vopson's primitive; bit budget = ΛCDM values as input, no forward content	—
UMF	Gericke	algebraic / Hilbert space	$\ell^2(P_N)$ is a proper subspace of $L^2(T^4)$; $\tau{\cdot}\mu = \hbar/c^2$ both ways	244
RA/PMT	Guevara	selection (meta-scheme)	FFGFT declines the selector slot RA1.5 (static)	245, 269
RSG	Austin	selection / bulk-boundary	asymmetric complementarity: FFGFT supplies scale, RSG selection language	246
UC	Haskins	constraint (static)	both static; α-correction inconsistency (H_0 vs. a_0)	260
UIFT	Teker	holographic boundary	observer-factor-3 is FFGFT's explicit version of the volume→boundary move	267, 268
αLGQV	Kriger	continuum-topological	different substrate type (S^3, membrane); not a lattice	—
Dot Theory	Vossen	meta / governance	orders the field (Lexicon/OAP/Matrix/FAH); no physical substrate	272

Lineage and external contacts (not IPI-forum frameworks, for completeness): *Scale Relativity* (Nottale) is the fractal-spacetime ancestor of FFGFT; the Koide circle (Koide, Rivero, Baez, Kocik, Brannen) concerns mass-formula verification, not the substrate.

.5 Why so often "a category difference, not a contradiction"

The map explains the recurring comparison result. FFGFT sits on *both* axes at the same structural-geometric pole: periodic crystal (axis 1) and geometry-primitive, static, no selector (axis 2). Most other frameworks build their theory precisely around a *slot* that FFGFT leaves empty in principle or fills differently:

- the **selector** (RA/PMT's RA1.5, RSG's selection principle) – FFGFT is static, nothing is selected;
- the **volume→boundary seam** (UIFT) – FFGFT makes it explicitly as observer-factor-3, but does not resolve the same open question;
- the **constraint** (UC) – FFGFT instantiates structure but formulates no standalone constraint principle;
- the **information primitive** (Vopson) – FFGFT takes geometry as primitive and derives information from it, not the reverse;
- the **external standpoint / simulation** (Vopson open) – structurally excluded in FFGFT.

Hence the findings are rarely contradictions: the frameworks speak about different layers. Where they genuinely *touch* – and can therefore collide empirically – is the substrate axis, and there the spectral discriminator (Doc. 271) provides the only currently sharp, shared test.

.6 Applying the governance layer: from Matrix entry to A*

In the governance vocabulary of Vossen (Dot Theory) and Guevara (RA/PMT) this map can be read not only as a static placement but as a *workflow*. The FFGFT–HLV axis is the concrete example – the workflow has already been run, not merely described:

- **Declared projection:** the substrate axis.
- **PRC** (Projection-Relative Classification): FFGFT periodic, HLV aperiodic.
- **PRM** (Projection-Relative Measurement): the spectral discriminator (level-spacing ratio $\langle r \rangle$, gap structure, spectral dimension) – a measurement defined *relative to that projection* and meaningless without it.
- **Residual** (declared, not absorbed): an independent reconstruction of the HLV spectrum ($\mathbb{Z}^6$ cut-and-project, k-NN Laplacian) shows that on the headline statistic $\langle r \rangle$ the quasicrystal is *not yet* separable from the random control; the aperiodic signature survives only weakly, in gap structure and the low-mode sector. This residual is declared, not computed away.
- **Operational accord:** the joint test protocol (periodic null, explicit recovery limit, scale-free ratios), bound by the two contributors and claiming no identity between the frameworks.

The resulting shared spectral testbed is a *Cooperative Admissible Space (A*)* in the sense of the recent Vossen–Guevara localisation: a declared region in which FFGFT and HLV cooperate *operationally* for a stated purpose without ontological identity. Guevara's observation holds: the accord is itself an object with its own admissibility conditions (the three agreed criteria), its own residual structure (what the matched run shows), and a revision operator (re-run once HLV supplies its weighted spectrum). It is *operational* rather than merely declared because the measurement can fail – the residual above is the proof that the space is doing real work. O0 and O2 are not the destination here but the instruments through which the accord becomes constructible and reviewable.

This section too is a declared application of others' vocabulary (Vossen, Guevara) and is subject to the originators' correction.

.7 Honest residue (declared)

- This map is a *declared reading* by the author (Dot-Theory O2). The primary rendering (O0) of each framework remains with its originator; the placements above are to be read as Matrix entries in the sense of Doc. 272, not as attributions.
- The *choice of axes itself* is a declared projection (Section 1), not a universal grid; other axes (Accessibility/Constraint, Observer-Dependence/Recoverability) are equally admissible and would yield a different legitimate map.
- The substrate triad (periodic/aperiodic/random) cleanly orders only FFGFT, HLV, and the control model. UMF (algebraic) and αLGQV (continuum-topological) sit beside the triad; their placement is via axis 2 or the substrate *type*, not the spectral fork.
- Placing Leavitt as "downstream of Vopson" follows from the provenance of his bit-budget construction (MEI), not from a standalone substrate claim.
- Should an originator operationalize their slot – e.g. Guevara casting RA1.5 as an equation, or Vopson stating an isolated, predicted, dimensionless ratio – the relevant row would be re-evaluated. The map is dated and provisional (June 2026).

References: Doc. 244 (UMF/Hilbert space), 245/269 (RA/PMT), 246 (RSG), 251 (Vopson), 260 (UC/Haskins), 267/268 (UIFT seam), 270 (projection chain $T^4 \to T^0$), 271/272 (HLV, pairwise and Dot governance), 190 R35 (ξ^N triviality). HLV: Krüger, Zenodo 10.5281/zenodo.20596861 and 10.5281/zenodo.20275888. Vopson: AIP Advances 2022/2023/2024. RA/PMT: Guevara 2026.